RC Networks and Synchronous Logic

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$$\dot{x} = \sigma(y - x),$$
$$\dot{y} = x(\rho - z) - y,$$
$$\dot{z} = xy - \beta z$$

σ=10, ρ=28, β=$\frac{8}{3}$

1 Objective and theory

Analyze a first-order RC network and document a synchronous register. All numeric results below are **illustrative calculations**, not laboratory measurements, simulation outputs, or FPGA test results.

$$H(s) = \frac{1}{1 + sRC}, \qquad \tau = RC, \qquad f_c = \frac{1}{2\pi RC}. \tag{1}$$

For $R = 1\,\mathrm{k\Omega}$ and $C = 100\,\mathrm{nF}$, $\tau = 100\,\mu\mathrm{s}$ and $f_c \approx 1.59\,\mathrm{kHz}$.

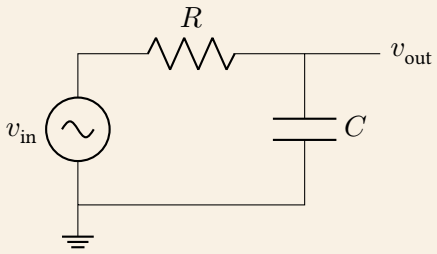


Figure 1: Low-pass RC circuit corresponding to equation (1).

2 Procedure and reproducibility

1. Record instrument models, calibration status, component tolerances, and the circuit revision before collecting data.
2. Sweep frequency across the predicted cutoff. Save raw data in `data/`; state amplitude conventions and probe loading.
3. Compare analytical predictions, simulations, and measurements in separate columns. Report uncertainty and explain discrepancies.

Table 1: Example result format; values are calculated placeholders.

Quantity	Predicted	Measured	Evidence status
τ (µs)	100	—	Measurement pending
f_c (kHz)	1.59	—	Measurement pending

3 Digital implementation and timing

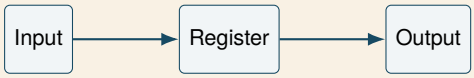


Figure 2: Minimal synchronous datapath.

```
module register8 (
    input  logic       clk,
    input  logic       rst,
    input  logic [7:0]  d,
    output logic [7:0]  q
);
  always_ff @(posedge clk) begin
    if (rst) q <= '0;
    else     q <= d;
  end
endmodule
```

Listing 1: Example register; compilation and hardware operation are not asserted.

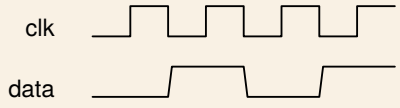


Figure 3: Illustrative clock and input waveform; not a simulator trace.

4 Results, limitations, and next steps

Distinguish RTL review, simulation, synthesis, timing analysis, and board testing. Record the exact tool

version, constraints, clock target, and evidence for every claim. For a timing report, state slack and corner conditions; for CPU or accelerator benchmarks, state workload, baseline, and measurement method.

Extensions.

- `plots`: frequency-response or measurement plots.
- `algorithms`: control pseudocode.
- `bibliography`: datasheet citations.